

13. Progetto e sviluppo di sistemi interattivi II - Waterfall e Agile

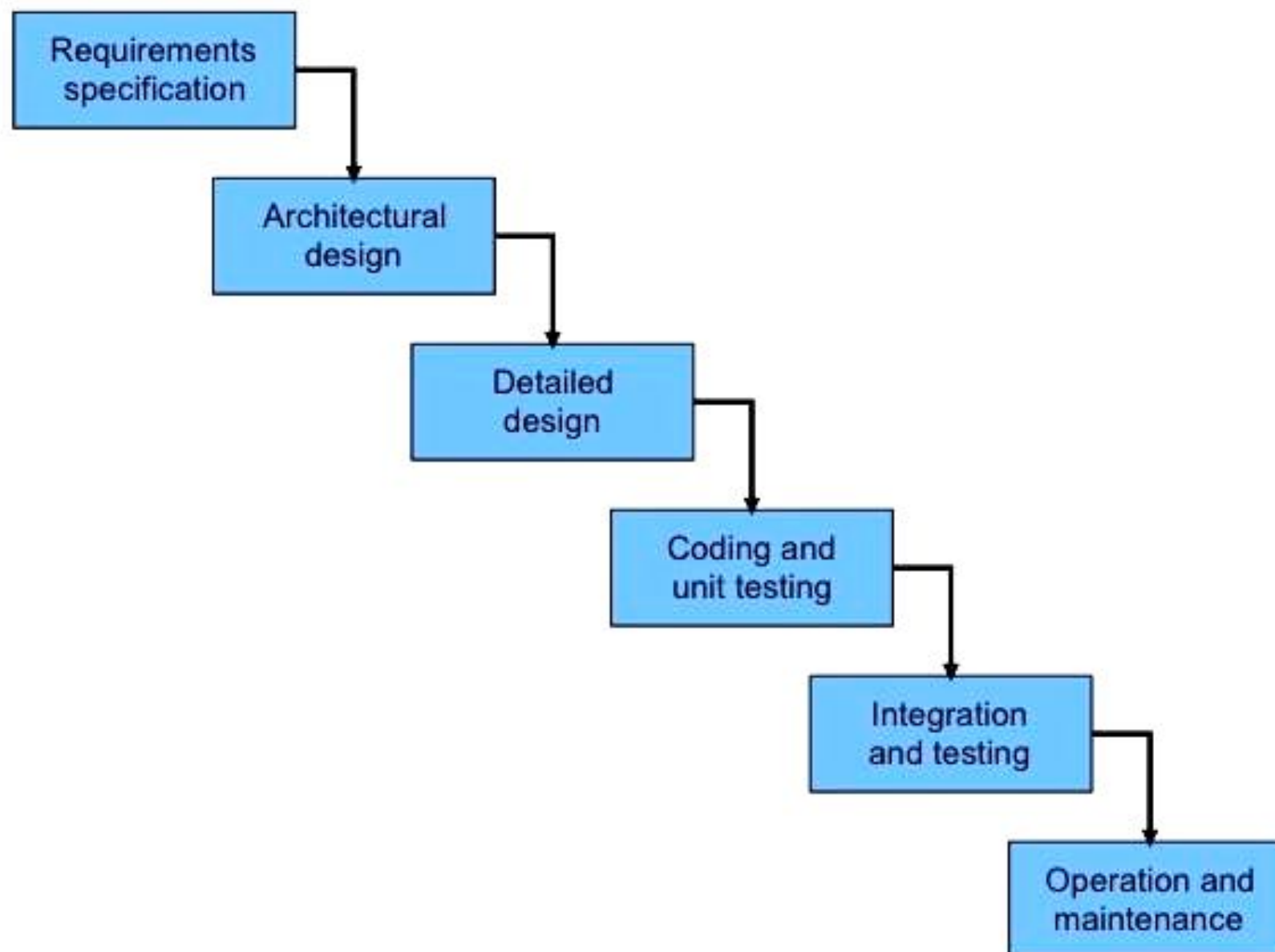
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The waterfall model



Activities in the life cycle

Requirements specification

designer and customer try capture what the system is expected to provide can be expressed in natural language or more precise languages, such as a task analysis would provide

Architectural design

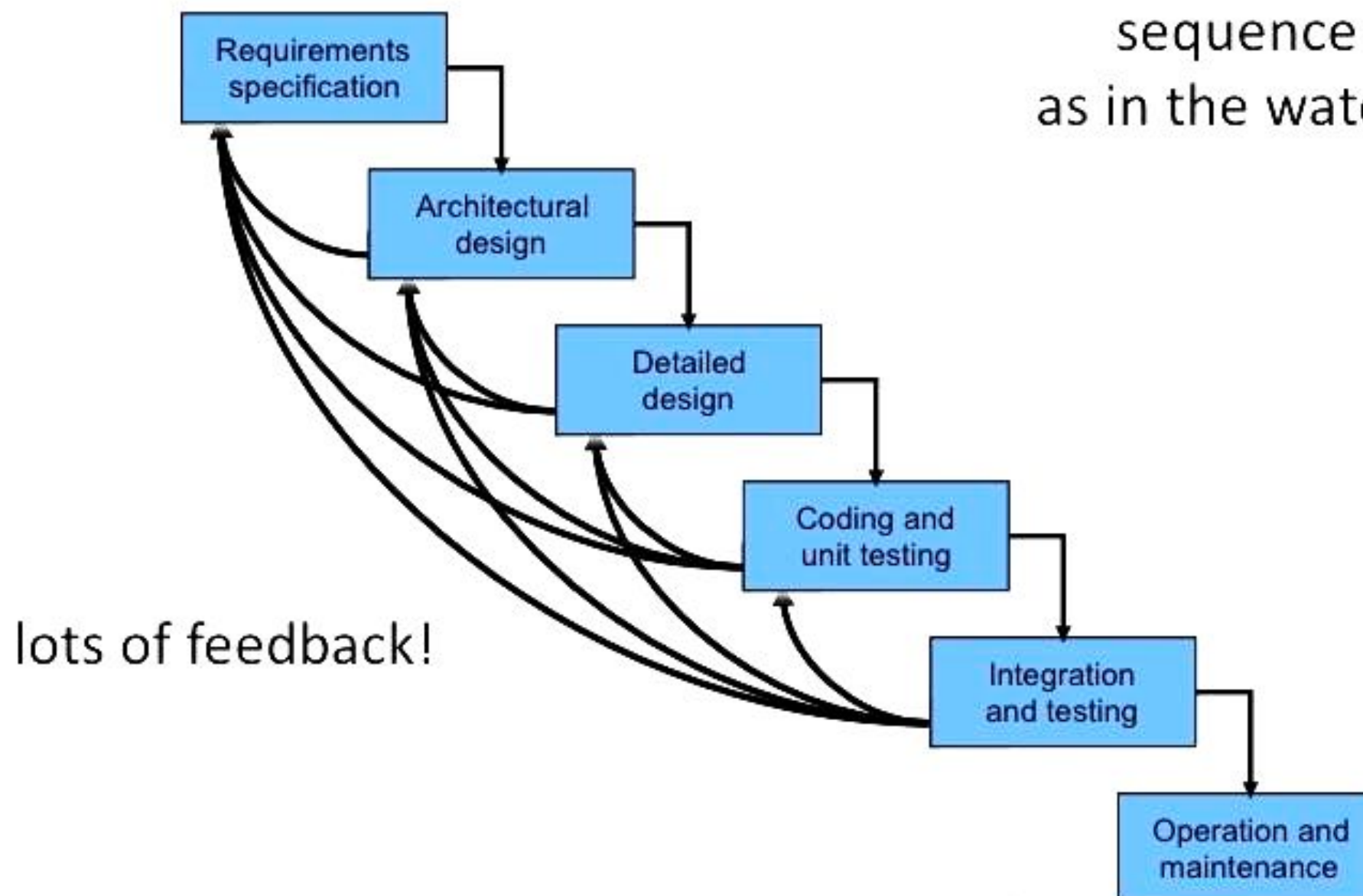
high-level description of how the system will provide the services required factor system into major components of the system and how they are interrelated needs to satisfy both functional and nonfunctional requirements

Detailed design

refinement of architectural components and interrelations to identify modules to be implemented separately the refinement is governed by the nonfunctional requirements

The life cycle for interactive systems

cannot assume a linear
sequence of activities
as in the waterfall model



Civil Engineering

- Design
 - difficult to predict
 - requires expensive and creative people
- Construction
 - easier to predict
 - less skilled intellectually, very skilled manually
 - much bigger in both cost and time than design and planning (90% vs 10%)

Software design&development

- Construction is so cheap as to be free
- All the effort is design, and thus requires creative and talented people
- Creative processes are not easily planned, predictability impossible

Waterfall often fails

- Separating design from development (for interactive sw systems) often fails because the design specifications change while developing:
 - Changing context
 - Better understanding of the problem from development
 - Discovery of unexpected usage
 - New business requirements
 - Technological base always advancing
 - New ideas

Iterative design and prototyping

- Iterative design overcomes inherent problems of incomplete requirements
- Prototypes
 - simulate or animate some features of intended system
 - different types of prototypes
 - throw-away
 - incremental
 - evolutionary
- Management issues
 - time
 - planning
 - non-functional features
 - contracts

Agile

- project management process that encourages
 - frequent inspection
 - adaptation,
 - teamwork, self-organization
 - rapid delivery of high-quality software,
 - align development with customer needs

Agile/2

- Agile methods are adaptive rather than predictive.
 - welcome change
 - even to the point of changing themselves.
- Agile methods are people-oriented rather than process-oriented.
 - the role of a process is to support the development team in their work.